



Hawler Medical University
College of Medicine, Medical Physics Unit

- Introduction about Medical Physics course
- Terminology, Modeling, and Measurement

(2 hours)

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PhD Medical Physics
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Terminology, Modeling, and Measurement

Objectives

- Understand the fundamentals of Medical Physics and its applications in diagnosing and treating diseases.
- Learn key terminologies used in Medical Physics and Medicine.
- Explore the role of physics in the human body and its medical applications, including various forms of electromagnetic radiation and diagnostic instruments.

Recommended text book:

- 1. Medical Physics for Medical students (Taki AM Almusa) 2024**
- 2. Medical physics (John R. Cameron) 1978 ,1993,1999,2003 and 2010**
- 3. Introduction of Health Physics (Herman Cember) 2009**
- 4. Introduction to physics in modern medicine (Suzanne A. kane) 2010**
- 5. Electronics in medicine and biomedical instrumentation (Nandini K. Jog) 2006**
- 6. Radiation protection and dosimetry (Michel G. Stabin) 2012**
- 7. Dr.R.N.Roy .Atext book of Biophysics 1st edition,2005**
- 8- Physics of radiotherapy Khan 2012**
- 9- Advance in Medical Physics, Devon J. Godfrey, Volume 6 2016**

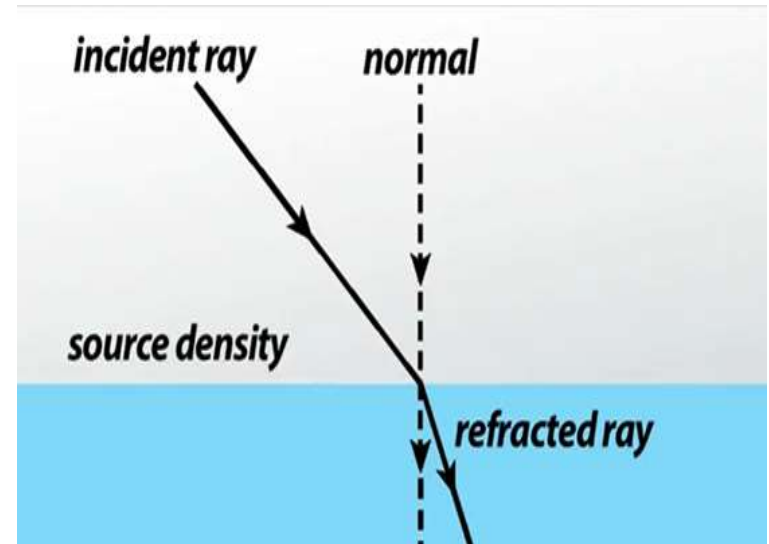
Terminology

1- Medical physics refers to two major areas: -

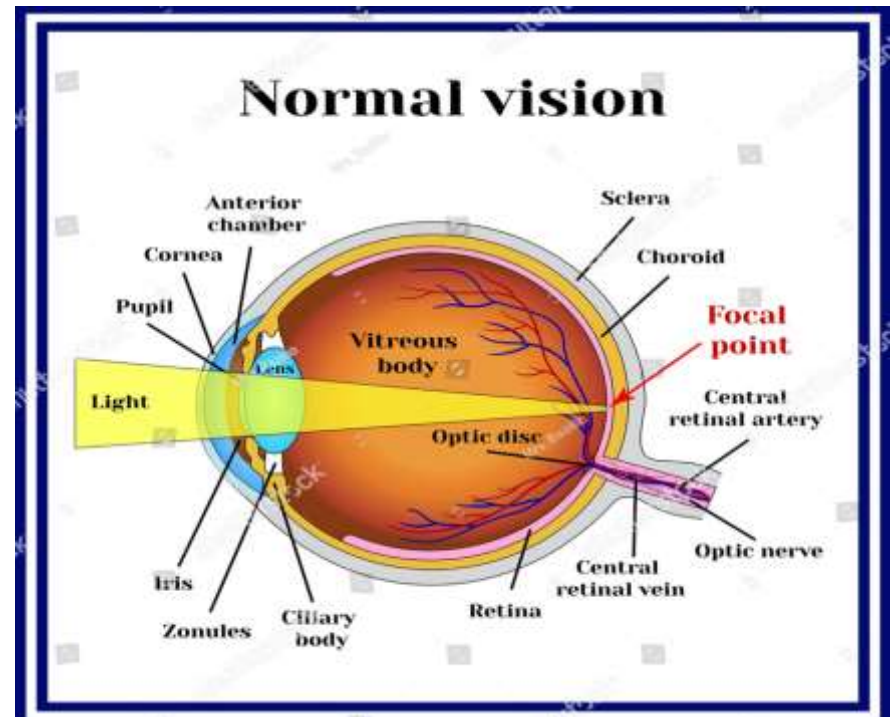
A. First area is The applications of physics to the function of the human body in health and disease (the physics of physiology).

Refraction of the light

When the light travels From a less dense to a denser medium (from air to the corneal), light refracts towards the normal (an imaginary line perpendicular to the surface at the point of entry).



In normal vision: Refraction in the eye is the bending of light rays as they pass through the cornea and lens to focus directly onto the retina, creating clear images.

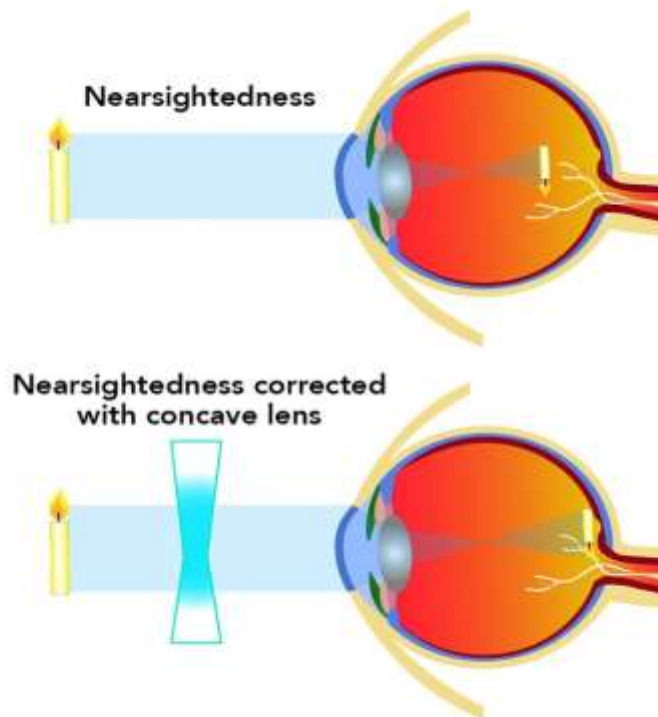


If the eye is unable to focus light correctly onto the retina, it can lead to a **refractive error** and vision problems.

Two common refractive errors are nearsightedness and farsightedness.

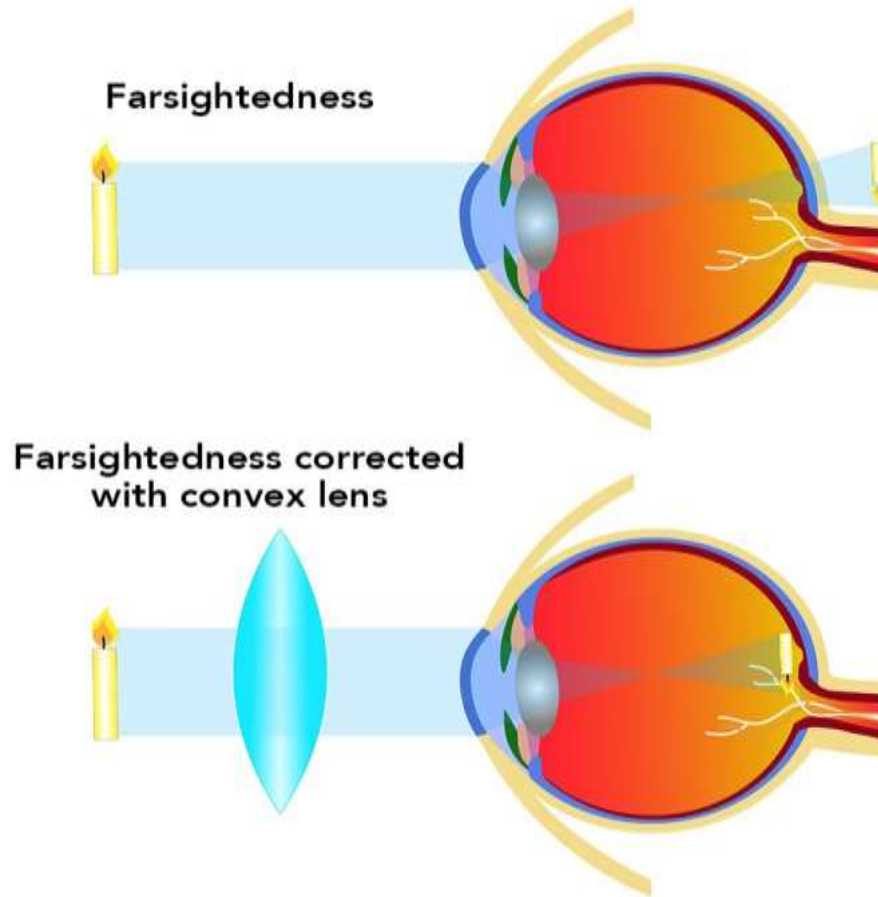
Individuals who are **nearsighted** or myopic can see objects up close, but objects at a distance appear blurry. Nearsightedness usually **develops in childhood** and stabilizes between ages **20 and 40**

The solution for **nearsightedness** according to physics law by utilizing concave lenses that gently diverge the light rays as they enter the eye.



Farsightedness (Hyperopia): Distant objects are clear, but near objects are blurry. Corrected according to physics law with converging (convex) lenses, marked with a plus sign (+).

This is among the most common problems adults develop between ages 41 to 60.



2- Second area of Medical physics

The applications of physics in the practice of medicine. This includes such things as the physics of the stethoscope, the tapping of the chest (percussion), and the medical applications of lasers, ultrasound, radiation and so forth.

Types of Electro Magnetic radiation (EMR)

radio waves

Microwaves

Shortwave

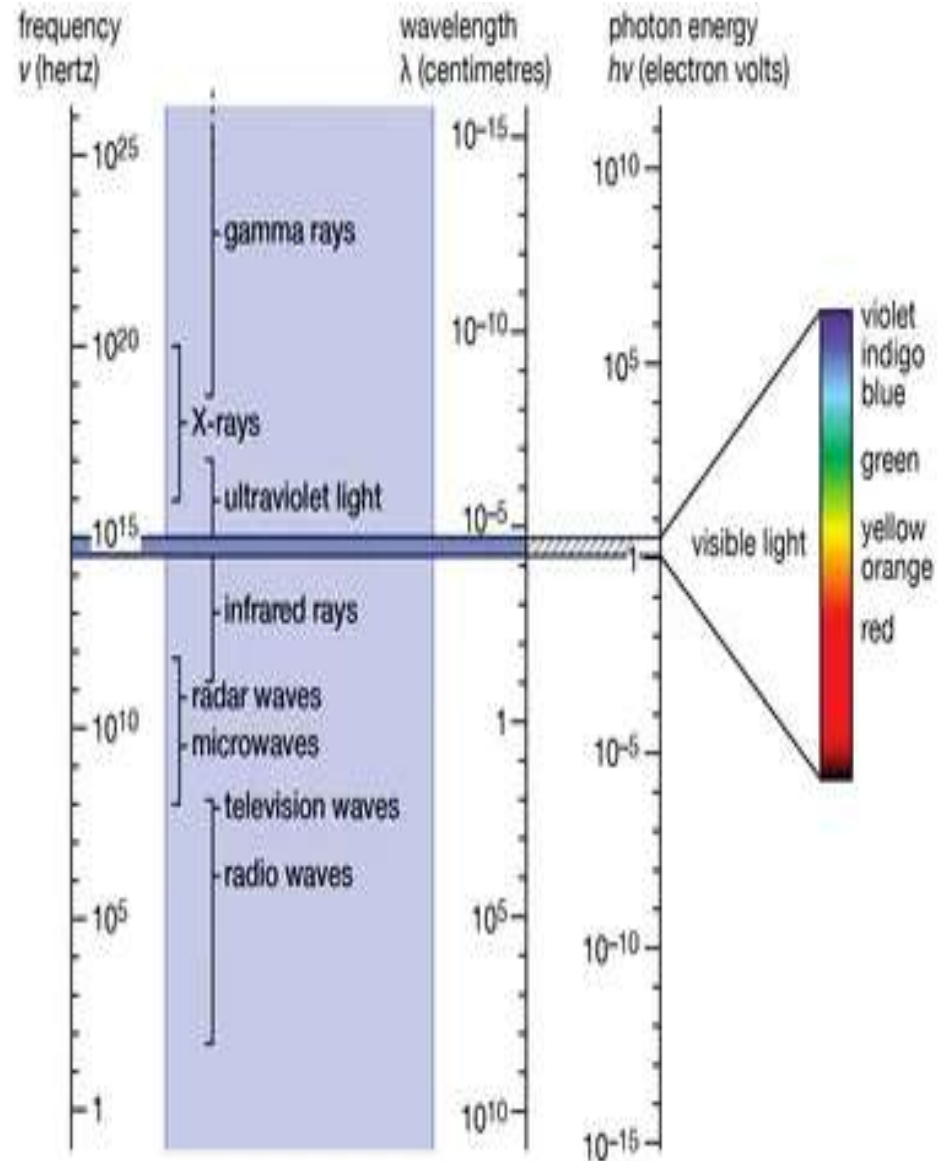
infrared radiation

visible light includes **LASER**

ultraviolet radiation

X-rays

gamma rays



Nonionizing

Ultraviolet, visible, infrared, microwaves,
radio & TV, power transmission

Ionizing

Radiation capable for producing ions
when interacting with matter – x-rays,
alpha, beta, gamma, cosmic rays

The use of magnetic and electromagnetic fields (EMF) in physical therapy is considered as an effective method of treatment of various diseases of the musculoskeletal system.

In physiotherapy, these electromagnetic modalities are generally used to expedite recovery of soft tissue injuries or relieve the pain. The main methods of therapy used for that purpose are: electrotherapy, magnetic therapy and phototherapy.

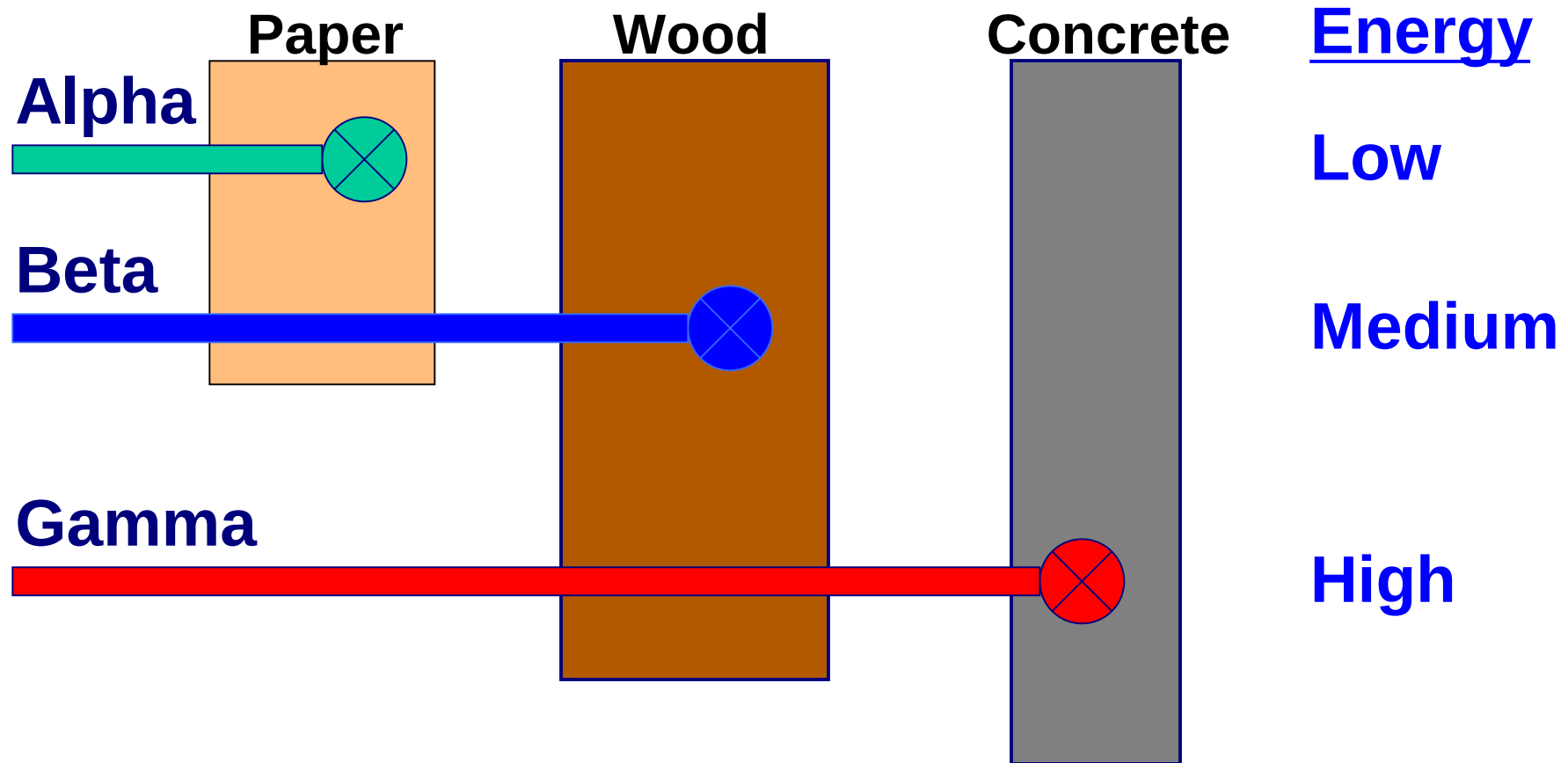
Ionizing Radiation

Ionization Defined

Radiation capable for producing ions when interacting with matter – in other words enough energy to remove an electron from an atom.

Sources – x-rays, radioactive material produce alpha, beta, and gamma radiation, cosmic rays from the sun and space.

Ionizing Radiation



Ionizing radiation is used in medicine in 3 ways:

- 1. Diagnostic radiology, which uses x-ray machines to obtain images of the inside of the patient's body.**
- 2. Radiation therapy uses high-energy particles or waves, such as x-rays, gamma rays, electron beams, or protons, to destroy or damage cancer cells.**
- 3. Nuclear medicine, which uses radioactive substances introduced into the patient for diagnosis or treatment.**

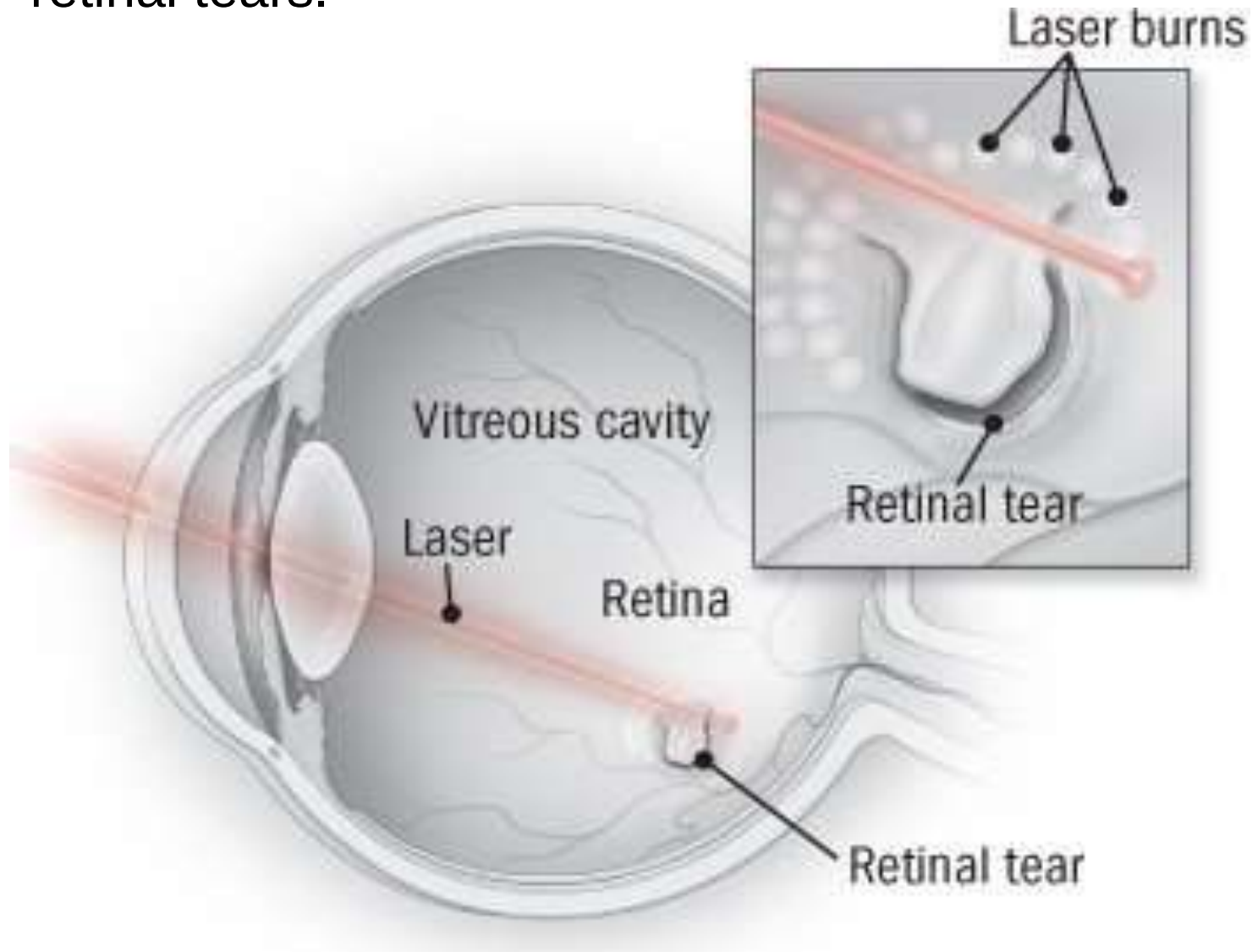
LASER In MEDICINE

LASER is (visible light) an acronym for light amplification by stimulated emission of radiation. It is type of EMR

It is focused in a narrow beam and creates a very high-intensity light.



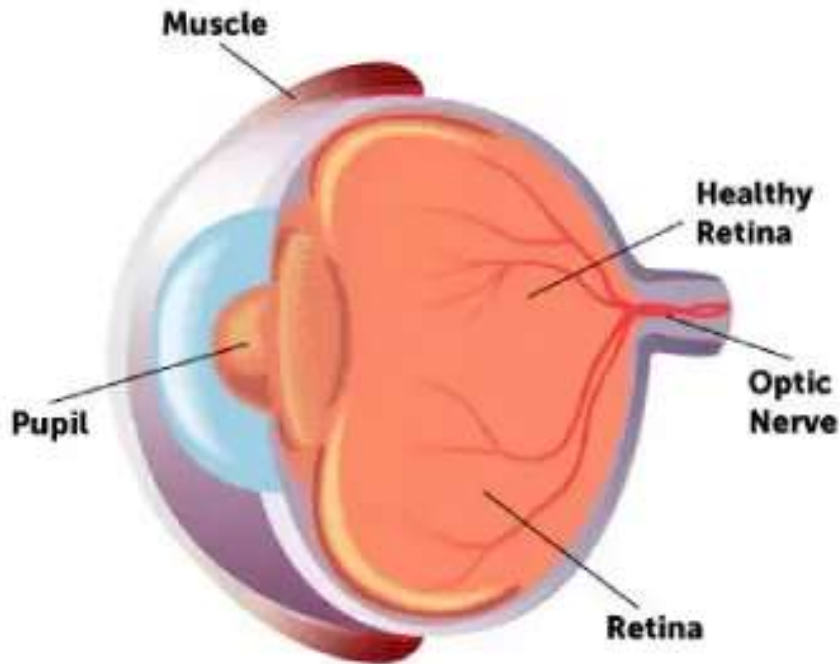
Lasers are used for photocoagulation of the retina to halt retinal hemorrhaging and for the tacking of retinal tears.



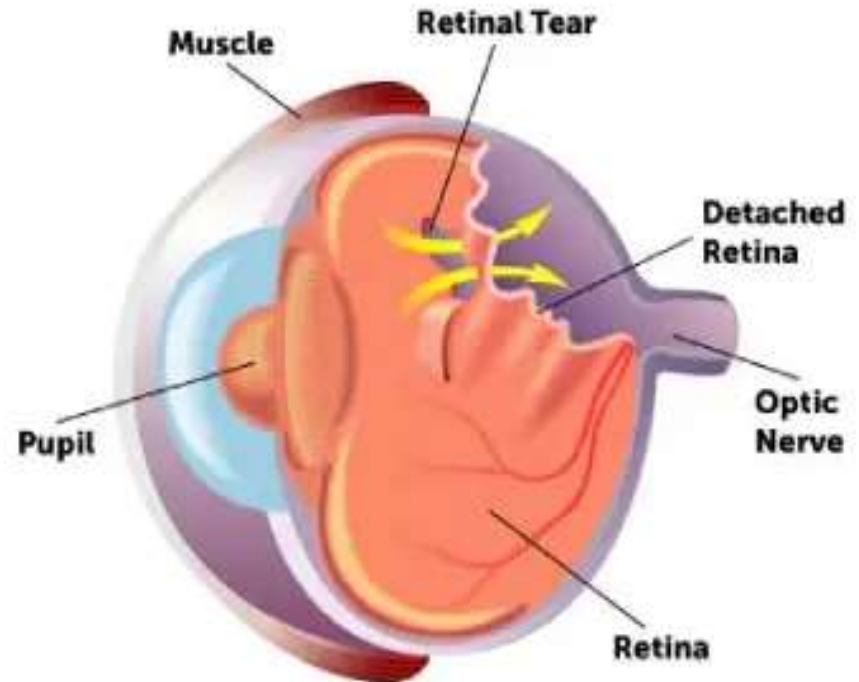
LASER is used to treating detached retinas

Retinal detachment happens when the thin layer of tissue at the back of the eye (Retina) pulls away from its regular position. Surgeon directs a laser beam into the eye through the pupil. The laser makes burns around the retinal tear to create scarring that usually "welds" the retina to underlying tissue.

HEALTHY RETINA

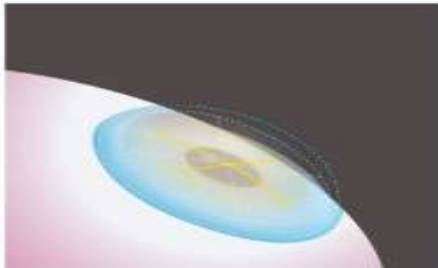


DETACHED RETINA

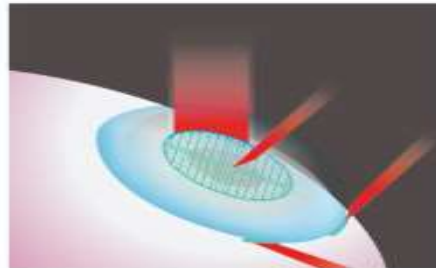


Laser Cataract Surgery

Laser cataract surgery is when an eye surgeon removes a clouded lens (the natural lens) with a laser, then replaces it with an artificial prescription lens that restores clear vision. This method results in better healing and fewer complications than traditional cataract surgery, which uses a scalpel.



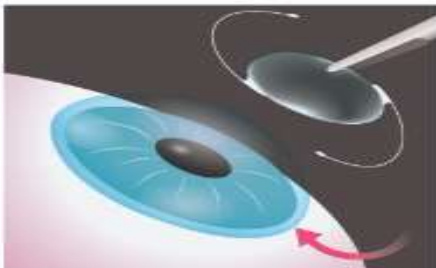
① Computer scan diameter lens.



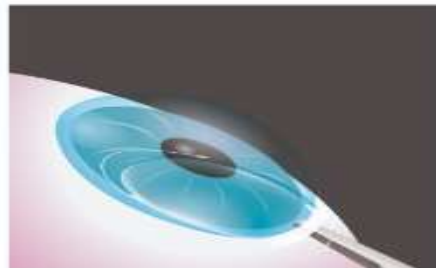
② Use femtosecond laser for cutting around lens in pieces, and cutting open for small tool inserts.



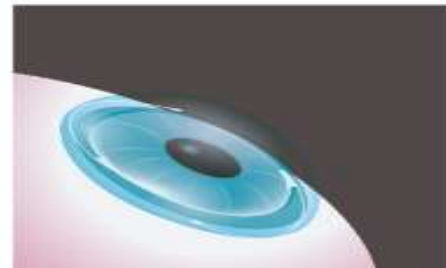
③ Use small rake and suction tool remove scrap lens.



④ Artificial intraocular lens (IOL) is placed in eye.



⑤ Insert artificial intraocular lens (IOL) in the opens tool slots.



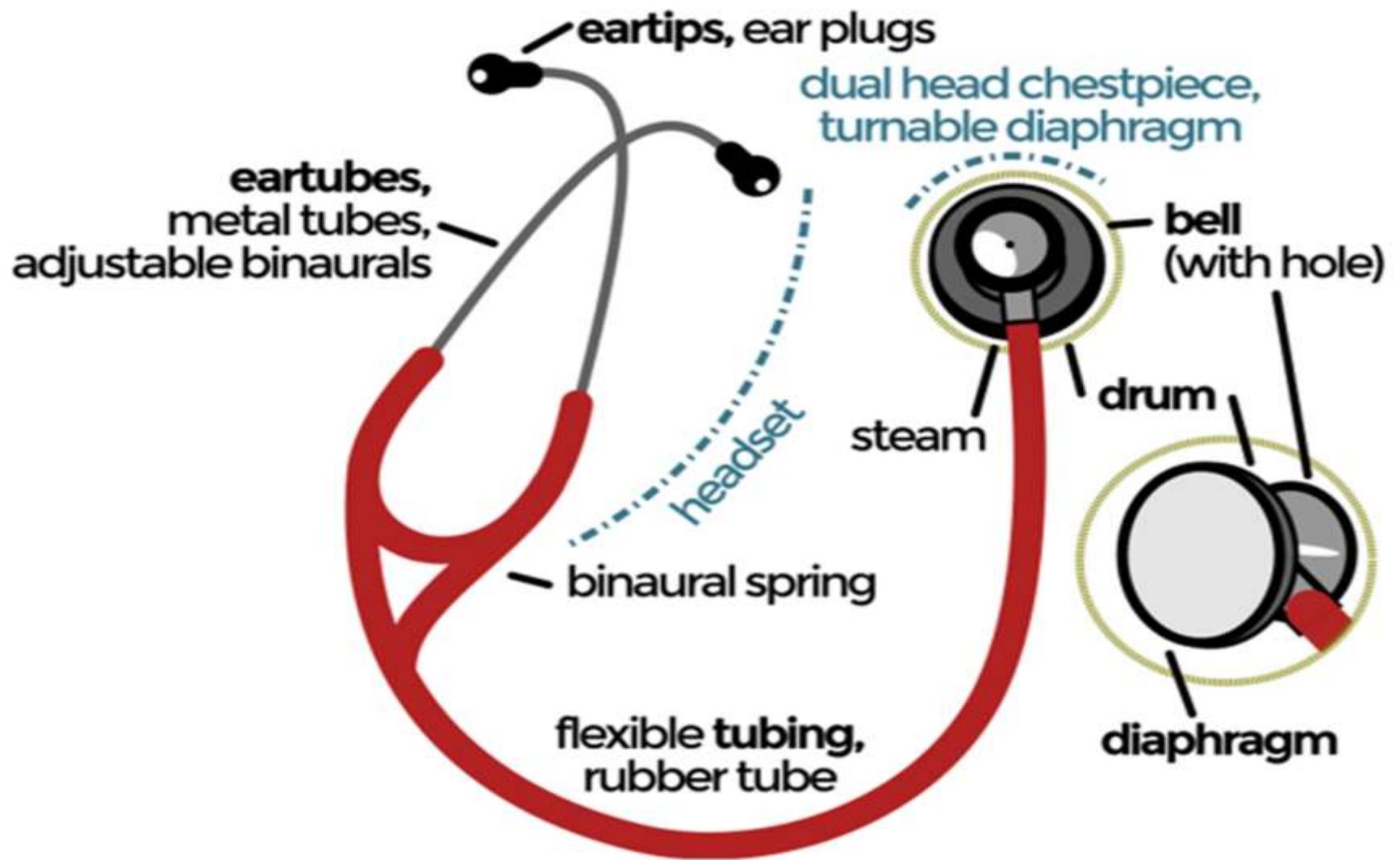
⑥ Artificial intraocular lens (IOL) into position. The surgery is finish.



In this photo taken during open-heart surgery, a doctor uses a laser probe to punch small holes in the patient's heart muscle to increase the organ's blood flow.

The physics of the stethoscope

Acoustic stethoscopes operate on the transmission of sound from the chest piece, via air-filled hollow tubes, to the listener's ears. The chestpiece usually consists of two sides that can be placed against the patient for sensing sound; a diaphragm (plastic disc) or bell (hollow cup).



If the diaphragm is placed on the patient, what is happened?



- If **the diaphragm** is placed on the patient, body sounds vibrate the diaphragm, creating acoustic pressure waves which travel up the tubing to the listener's ears
- **The diaphragm transmits higher frequency sounds.** The disc and the tube of the stethoscope amplify small sounds

If the bell is placed on the patient, what is happened?



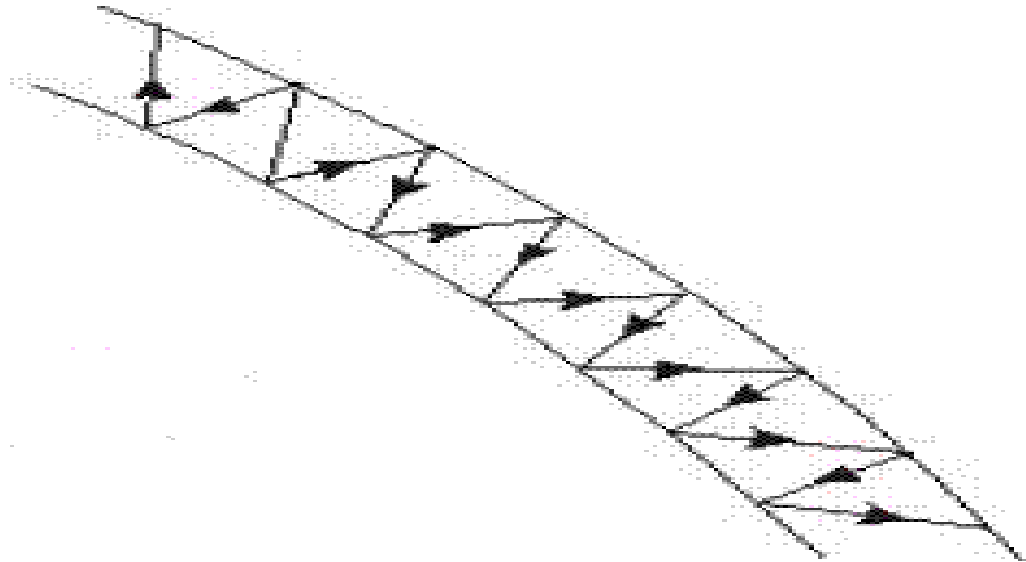
If the **bell** is placed on the patient, the vibrations of the skin directly produce acoustic pressure waves traveling up to the listener's ears.

The bell transmits **low frequency** sounds, while the diaphragm transmits higher frequency sounds.





Stethoscope



**Multiple reflection of sound
in the tube of stethoscope**

The disc and the tube of the stethoscope amplify small sounds by having a large surface to collect the sound, then channeling it to the small tubes that lead to the listener's ears.

Multiple Reflections: Sound waves travel through the air inside the tube and undergo multiple reflections off the inner walls. This guides the sound waves efficiently along the tube to the earpieces amplify with minimal loss of intensity.

The physics of percussion -

Percussion is an assessment technique which produces sounds by the examiner tapping on the patient's chest wall. So tapping on the chest wall produces sounds based on the amount of air in the lungs. Percussion sets the chest wall and underlying tissues into motion, producing audible sounds and palpable vibrations. Percussion helps to determine whether the underlying tissues are filled with air, fluid, or solid material.

Percussing the anterior chest is most easily done with the patient lying supine; the patient should sit when percussing the posterior chest. Place the first part of the middle finger of your non dominant hand firmly on the patient's skin. Then, strike the finger placed on the patient's skin with the end of the middle finger of your dominant hand



Physical medicine

The branch of medicine referred to as physical medicine deals with the diagnosis and treatment of disease and injury by means of physical agents such as manipulation, massage, exercise, heat, and water.

Physical therapy is the treatment of disease or bodily weakness by physical means such as massage and gymnastic rather than by drugs.

The field of medical physics has several subdivisions

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graph TD; A[The field of medical physics has several subdivisions] --> B[Diagnostic radiology]; A --> C[Nuclear medicine]; A --> D[radiation therapy]; A --> E[Health physics];
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Diagnostic radiology

Which includes X-ray, computed tomography (CT scan), Ultrasound US and magnetic resonance image (MRI).

Nuclear medicine

Which deals with the use of radioactive substances in diagnosis, treatment and research.

radiation therapy

its medical specialty that involve treating cancer by radiation

Health physics

is the application of scientific principles to the protection of worker and public from the hazards of radiation.

Modeling

The physicists believe that the physical world obeys the laws of physics.

Physicists often teach and think by analogy.
Therefore Analogies employ as models.

Analogies is used to understand the physical aspects of the body. (Not perfect), such as the eye analogous to camera.

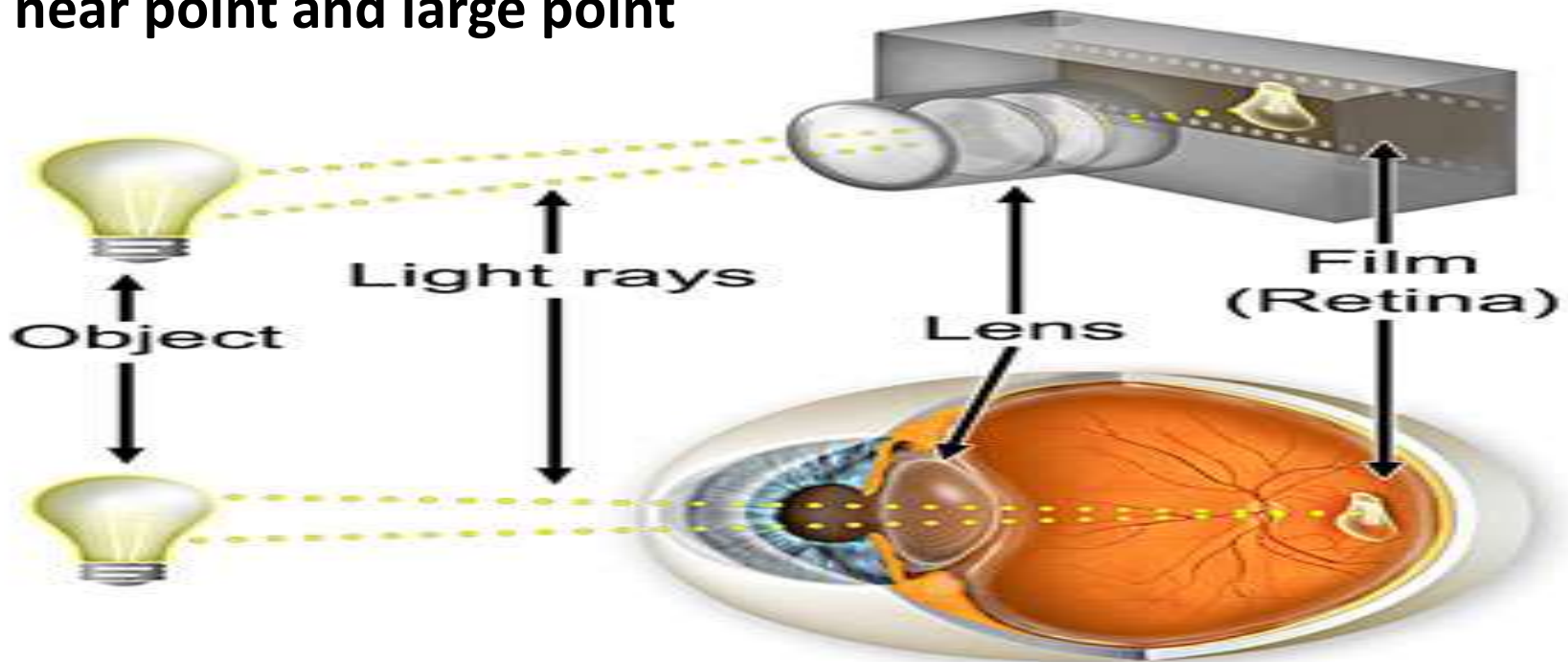
To describe and explain phenomena that can't experience directly physicists employ models, using different types for different aims.

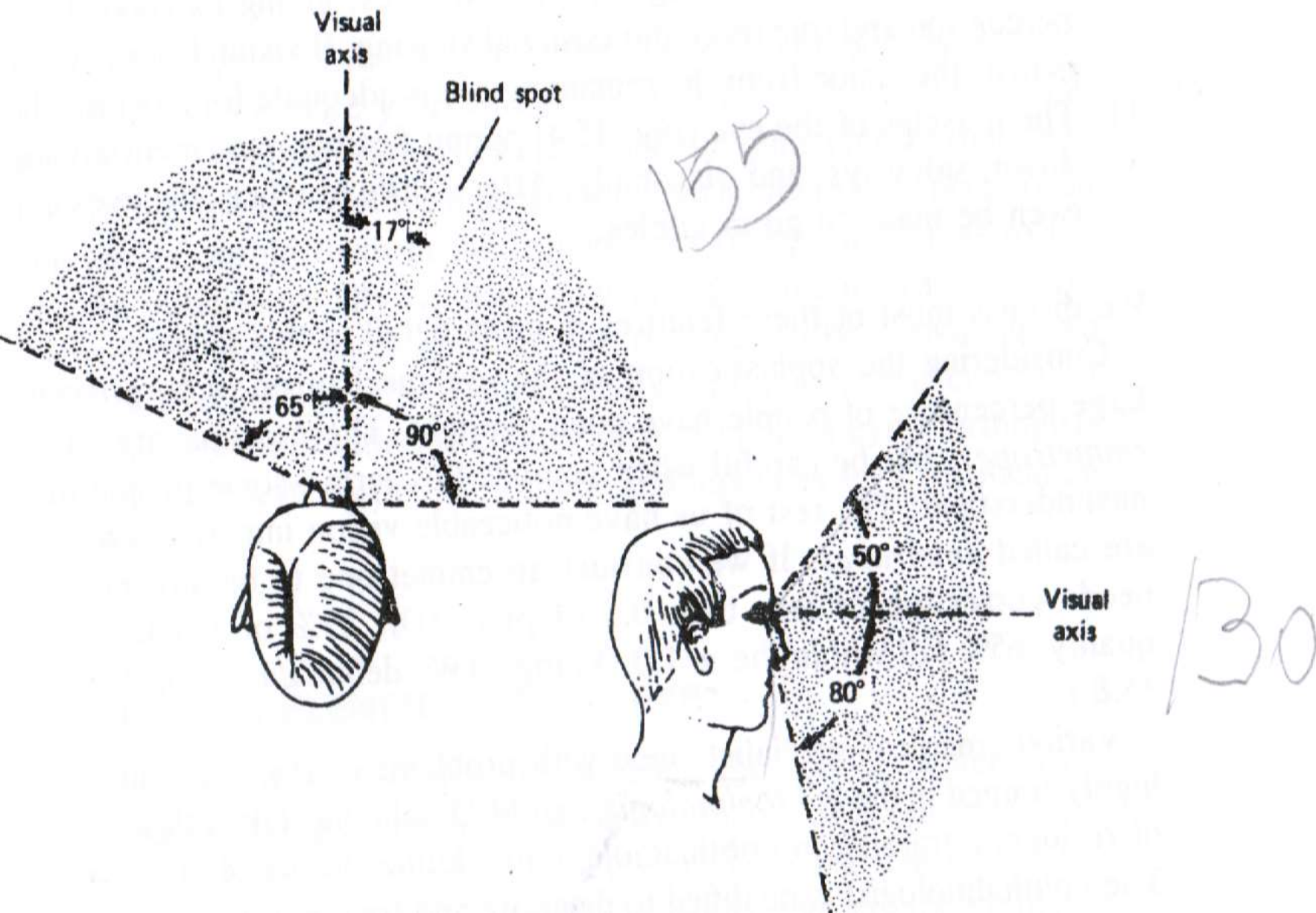
First Example : -

In many ways the eye is analogous to a camera.

This analogies are never perfect, because our optical system has special features are not available on the camera.

- The eye can observe event over large angle
- Blinking provides cleaning and lubrication to front lens
- A rapid automatic focusing system permits viewing object to near point and large point





The previous model involves **physical phenomena**.
Other models are **mathematical**.

Equations are mathematical models that
can be used to describe and predict the
physical behavior of some systems

Ex.: the relationship between Heart rate R is a
function of the power produced by the body P

$$R = f(P)$$

Measurements involving the body and time can be divided into two groups: -

- 1. Measurements of repetitive processes**
- 2. Measurements of non repetitive processes,**

Measurements of the repetitive processes
usually involve the number of repetitions per second, minute, hour, and so forth.

For example: -

- The pulse rate is about 70/min.
- The breathing rate is about 15/min.

Measurements of non repetitive processes
such as how long it takes the kidneys to remove a foreign substance from the blood.

When a physician must decide if the patient is ill or not, and what the illness is?

After a physician has reviewed a patient's: -

1. Medical history.
2. The findings of the physical examination.
3. The results of the clinical laboratory measurements

The decisions are two types: -

1. Right decisions.
2. Wrong decisions.

It is not surprising that sometimes wrong decisions are made. These wrong decisions are of two types: -

1. False Positives.

2. False Negatives

A false positive error occurs when a patient is diagnosed to have a particular disease when he or she does not have it.

A false negative error occurs when a patient is diagnosed to be free of a particular disease when he or she does have it.

Diagnostic errors (false positives and false negatives) can be reduced by: -

1. Research into the causes of misleading laboratory test values.
2. Development of new clinical tests and better instrumentation.

Errors from measurements can be reduced by: -

1. Using care in taking the measurement.
2. Repeating measurements.
3. Using reliable instruments.
4. Properly calibrating the instruments.

Questions

1. What is the primary use of lasers in ophthalmology?
2. What does the diaphragm of a stethoscope do?
3. What type of error occurs when a patient is incorrectly diagnosed with a disease they do not have?
4. Which of the following is NOT a type of electromagnetic radiation?
 - a. Gamma rays
 - b. Ultraviolet radiation
 - c. Alpha particles
 - d. Microwaves
5. What is the main advantage of laser cataract surgery over traditional methods?
6. What role does percussion play in a physical examination?
7. Explain the significance of using models in physics to understand human physiology.
8. What are two methods to reduce diagnostic errors in medicine?
9. What is the main function of the bell side of a stethoscope?
10. What is the main purpose of the bell side of a stethoscope during a physical examination?
11. What type of sounds does the bell side of a stethoscope primarily transmit?
12. In general, electromagnetic radiation is divided into two groups. In which medical field is each group used?
13. What steps or examinations does the doctor rely on to make the final decision regarding the patient?

THANKS....